

# Sadhana Jamkatel

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## EDUCATION

**North Central College**, Naperville, IL

*Expected: May 2027*

**Major:** Bachelor of Science in Computer Science, **Minor:** Economics

**GPA 3.87/4**

**Relevant Coursework:** Data Structures, Computer Organization/Design, Software Development, Database System, Algorithm & Design, Computer Network and Security, Algorithms and Design

## WORK EXPERIENCE

**DevOps Engineer Intern**

*Jun 2025-Aug 2025*

*Aeropay*

- Reduced documentation search time from 30 minutes to 30 seconds by building a full-stack internal documentation site using Docusaurus (MDX, custom charts, integrated search), serving 30+ engineers daily and hosted on GitHub Pages with automated CI/CD via GitHub Actions ; also tested the site using AWS S3 + CloudFront with JumpCloud SSO access controls
- Created a documentation site for a CLI tool (used for automating login, role-based access, and token generation for AWS and databases) that auto-generates and updates files using OCLIF's markdown generator and GitHub Actions.

**High-Performance Computing (HPC) Bootcamp Participant**

*Aug 2025*

*Argonne National Laboratory*

- Developed HPC chatbot for scientific Q&A using LangChain, HuggingFace, and Gemini API with RAG architecture, achieving 35% accuracy improvement and 45% latency reduction (3.2s to 1.8s) compared to non-RAG baseline

## PROJECTS

**NVIDIA Jetson AGX Thor Object Detection (Python, OpenCV, YOLOv8)**

*Jan 2026- Present*

- Implemented an embedded computer vision system on NVIDIA Jetson AGX Thor using Python, OpenCV, and YOLOv8 to process real-time camera input, preparing the system for automated dataset analysis
- Configured headless Linux environment, virtual environments, and project workflow for efficient development, testing, and reproducible results.
- Validated object detection on live and pre-recorded inputs, and automated data transfer via SCP, laying groundwork for future model training and performance evaluation.

**Care Equity | Full-Stack Healthcare Bias Tracker**

*Feb 2026*

*Hackathon Code2040*

- Engineered a full-stack Next.js and Node.js application to surface maternal healthcare inequities, featuring a responsive hospital discovery tool, interactive data visualizations (Recharts), and an anonymous reporting system.
- Designed a robust REST API with Express and MongoDB, implementing a custom same-origin API proxy to resolve cross-browser and mobile compatibility issues, ensuring seamless data delivery across all devices.
- Managed end-to-end deployment using Vercel (Frontend) and Render (Backend), configuring environment-based routing and automated "keep-awake" strategies to maintain high availability for production users.

**Predictive Maintenance ML System for Fleet Engine Monitoring**

*Sep 2025*

- Reduced engine downtime by 95% by building a predictive maintenance system using Random Forest to analyze 21 sensor readings from NASA Turbofan data and detect at-risk engines before failure.
- Designed React/Typescript dashboard with alerts and risk visualization to monitor 100 engines and notify maintenance teams of critical issues.
- Achieved 96% accuracy with 4.28% false alarm rate, delivering \$750K+ projected annual cost savings through optimized hyperparameter tuning and feature engineering.

## ON CAMPUS INVOLVEMENT

**Resident Assistant | North Central College**

*May 2024 – Present*

- Increased engagement by 80% by organizing 15+ events per semester and providing mentorship to 100+ residents

**Vice President | Society of Hispanic Professional Engineers (SHPE)**

*Aug 2025 – Present*

- Secured conference funding and led career development workshops for engineering students while overseeing executive board operations and coordinating semester programming system

**President | Computer Science Club**

*May 2023 – Present*

- Grew club to 25+ members by organizing 10+ technical workshops with industry professionals, leading collaborative projects, and fostering peer mentorship to increase retention and hands-on learning.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, C/C++, Java

**Web & Frameworks:** React.js, Node.js, Flask, Next.js, HTML/CSS

**Cloud & DevOps:** AWS S3, EC2, CloudFront, CloudWatch, Docker, GitHub Actions, Linux/Unix, Kubernetes, Salesforce Sales Cloud, Salesforce CRM

**Data & ML:** PostgreSQL, MongoDB, scikit-learn, pandas, NumPy, LangChain, SQL, GenAI tools, Yolo, OpenCV